

A Survey on Part-of-Speech Tagging

for Twitter / Social Media Data

Undergraduate Project — Literature Survey

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Abstract

Part-of-speech (POS) tagging is one of the foundational tasks in natural language processing (NLP), providing morpho-syntactic labels to words that serve as input to a wide range of downstream applications. Traditional POS taggers were trained predominantly on newswire corpora such as the Penn Treebank and perform reliably on well-formed formal text. However, Twitter and other social media platforms present a radically different linguistic environment: text is short, noisy, abbreviation-heavy, laden with novel entities such as hashtags and user mentions, and often grammatically unconventional.

Journalistic tweets introduce an additional and under-studied layer of complexity—dropped determiners, emoji functioning as syntactic units (e.g., an alert emoji used in place of the word *urgent*), compound hashtags encoding named entities, and domain-specific markers such as **BREAKING:** and **RT**—that neither general-purpose nor generic social media taggers are architecturally equipped to handle. Existing approaches force a binary choice between deep syntactic understanding (standard BERT) and Twitter domain familiarity (BERTweet), and none natively inject structural token-type metadata into the transformer’s representational pathway.

This survey reviews the key challenges of tweet tokenisation and POS tagging with a particular focus on the journalistic subdomain, examines the landmark datasets and annotation schemes proposed for Twitter text, surveys rule-based, statistical, and neural approaches developed to address those challenges, and identifies this structural metadata gap as the central open problem motivating the design of a news-aware tagger.

1 Introduction

The emergence of social media as a primary channel for public communication has created enormous opportunities for computational analysis of real-world language use. Twitter generates hundreds of millions of short messages daily, covering topics ranging from breaking news to personal expression. Mining this data for sentiment, opinion, events, named entities, and trends requires linguistic pre-processing—and POS tagging sits at the very beginning of that pipeline.

POS tagging assigns a grammatical category—noun, verb, adjective, and so on—to each token in a sequence. These labels encode syntactic and semantic information that downstream tasks such as named entity recognition (NER), dependency parsing, sentiment analysis, and information extraction all depend on. The accuracy of a POS tagger has a compounding effect: errors propagate through the entire processing chain.

The challenge with Twitter text is well documented. Gimpel et al. (2011) were among the first to systematically quantify the gap between standard tagger performance on newswire and on tweets, reporting an accuracy drop of over 10 percentage points for the Stanford tagger when applied directly to tweet data. Since then, the community has responded with specialised annotation schemes, new training corpora, domain-adapted statistical models, and deep neural architectures capable of learning rich contextual representations.

However, a specific and practically important subdomain remains largely unaddressed: *journalistic tweets* from breaking-news accounts. These tweets exhibit a distinct linguistic register characterised by telegraphic compression, heavy use of capitalised alert markers (BREAKING:, JUST IN:), emoji deployed as functional syntactic units, and compound hashtags encoding proper nouns and event names. Off-the-shelf taggers—including BERTweet, the strongest generic tweet tagger available—fail on this register because they treat all tokens as semantically flat text, with no architectural mechanism to exploit structural token-type information.

This survey is structured as follows. Section 2 characterises the properties of Twitter language that make it difficult to process. Section 3 reviews existing datasets and annotation schemes. Section 4 surveys tagging approaches, culminating in domain-adapted Transformers and the structural metadata gap. Section 5 covers evaluation methodology. Section 6 presents a comparative analysis. Section 7 identifies open challenges. Section 8 concludes.

2 Characteristics of Twitter Language

2.1 Structural Constraints and Their Linguistic Consequences

Twitter originally imposed a strict 140-character limit (expanded to 280 characters in 2017). This constraint incentivises extreme compression, driving users towards a dense shorthand that violates virtually every assumption made by taggers trained on formal text.

2.1.1 Tokenisation Challenges

Standard tokenisers assume whitespace-delimited tokens and punctuation as sentence boundaries. Tweets routinely violate both. Hashtags such as **#StockMarket** concatenate multiple morphemes without spaces—a compound a standard morphological analyser treats as a single unknown token rather than a bigram of proper nouns. Emoji function as lexical items with no alphabetic representation, yet carry clear syntactic weight: an alert emoji at the start of a journalistic tweet occupies the structural role of an interjection or discourse marker. O’Connor et al. (2010) proposed the *Ttokenizer*, one of the first tokenisers designed for Twitter, recognising the need to handle these tweet-specific token types explicitly.

2.1.2 Orthographic Variation

Tweets exhibit pervasive creative orthography. A single word may appear in multiple forms: *tomorrow*, *tmrw*, *tmr*, *tomoro*. Letter casing is inconsistently used—ALL CAPS can signal shouting, irony, or, in the journalistic domain, a headline-style alert (**BREAKING**, **EXCLUSIVE**). Standard morphological analysers and lookup-based taggers break down under such variation.

2.1.3 Twitter-Specific Token Classes

Twitter introduces token classes absent from traditional tagsets. The annotation scheme of Gimpel et al. (2011) includes dedicated tags for:

- **Hashtags** (**#topic**): discourse markers, metadata labels, and neologisms; in the news domain, often compound proper nouns such as **#SupremeCourt** or **#StockMarket**.
- **At-mentions** (**@user**): addressivity markers; in journalistic tweets, often source attributions (**@Reuters**, **@AP**).
- **URLs**: which may function as quoted material, evidence, or context.
- **Retweet markers** (**RT**): a pragmatic discourse device signalling reported speech.
- **Emoticons and emoji**: affective and illocutionary signals; in breaking-news tweets, functionally equivalent to alert words.

- **News-specific markers:** tokens such as `BREAKING:`, `JUST IN:`, `EXCLUSIVE:` that carry discourse-structuring force absent from general Twitter text.

Failing to recognise and correctly tag these classes leads both to direct tagging errors and downstream confusion. For a transformer-based tagger, the problem is not merely vocabulary coverage: even a model that has seen these tokens during pre-training may assign incorrect POS labels if it lacks any architectural signal distinguishing a hashtag from a regular noun, or an emoji from punctuation.

2.2 Lexical and Grammatical Non-Standardness

Twitter language includes high rates of slang, code-switching, and dialectal variation. African-American Vernacular English (AAVE), disproportionately present in Twitter corpora, involves morphosyntactic patterns—habitual *be*, zero copula, negative concord—that deviate from standard English patterns. Eisenstein (2013) discusses these sociolinguistic dimensions and their implications for NLP system design.

In the journalistic subdomain, the dominant non-standard pattern is *telegraphic compression*. Determiners are routinely dropped (*Police arrest suspect* rather than *The police have arrested a suspect*), subjects may be omitted, and sentences often lack a main verb. The result is a register that is simultaneously more formal in vocabulary and less formal in grammar than either newswire or casual social media text.

3 Datasets and Annotation Schemes

3.1 Gimpel et al. (2011) Dataset and Tagset

The most influential resource for English tweet POS tagging is the dataset introduced by Gimpel et al. (2011), distributed as part of the ARK Twitter NLP toolkit. The dataset comprises approximately 1,800 tweets (around 26,000 tokens), manually annotated with a 25-tag coarse-grained tagset. Beyond Penn Treebank-derived categories, the tagset adds tags for hashtags (`#`), at-mentions (`@`), URLs (`U`), emoticons (`E`), and a catch-all for other Twitter-specific tokens. Notably, the dataset was drawn from a general Twitter sample and does not include a dedicated journalistic subcorpus.

3.2 Owoputi et al. (2013) Extended Dataset

Owoputi et al. (2013) extended the Gimpel dataset with additional tweets and a revised annotation process, producing the benchmark used in most subsequent comparative evaluations. This dataset remains the de facto standard for evaluating English tweet POS

taggers and is used as the primary benchmark in this work.

3.3 Universal Dependencies for Twitter

The Universal Dependencies (UD) project provides cross-linguistically consistent annotation for dependency parsing and POS tagging. The TWEEBANK corpus (Liu et al., 2018) contains 1,639 tweets annotated with UD POS tags and dependency relations, facilitating multilingual comparison and transfer learning from high-resource newswire data.

3.4 Multilingual and Domain-Specific Corpora

Research beyond English has produced Twitter POS datasets for Spanish (Derczynski et al., 2013), French, German, and Arabic. To our knowledge, no publicly available dataset specifically targets the journalistic breaking-news register—a data gap that directly constrains progress in this area.

Table 1: Overview of major tweet POS tagging datasets.

Dataset	Lang.	Tokens	Tagset	Public
Gimpel et al. (2011)	English	26,000	25-tag Twitter	Yes (ARK)
Owoputi et al. (2013)	English	~28,000	25-tag (revised)	Yes (ARK)
TWEEBANK v2 (2018)	English	~28,000	UD POS + deps	Yes (GitHub)
TweetNLP Shared Task (2014)	Multi	Varies	Various	Partial
Derczynski et al. (2013)	English	~7,000	Penn + extensions	Yes

4 Approaches to Tweet POS Tagging

4.1 Rule-Based and Dictionary-Based Approaches

Early attempts applied off-the-shelf rule-based systems with Twitter-specific preprocessing. Han and Baldwin (2011) proposed a lexical normalisation system mapping non-standard tokens to standard equivalents before tagging. While interpretable and requiring no training data, these systems are brittle and do not cover journalistic markers or domain-specific emoji usage.

4.2 Statistical Sequence Labelling Models

4.2.1 Hidden Markov Models

HMMs dominated POS tagging before the neural era. For tweet tagging, HMMs trained on newswire data suffer severely from the unknown word problem: a large fraction of tweet tokens are out-of-vocabulary, and emission probabilities for unknown tokens require heavy smoothing. HMM-based baselines consistently underperform on social media text.

4.2.2 Conditional Random Fields

Conditional Random Fields (Lafferty et al., 2001) model the conditional probability of the tag sequence given the observed word sequence, allowing arbitrary overlapping features. Gimpel et al. (2011) trained a CRF using lexical features, character n -grams, token shape features, Twitter-specific indicator features, and Brown cluster features derived from 56 million unlabelled tweets. Brown clusters were critical for generalising across novel surface forms.

Owoputi et al. (2013) further improved this baseline by expanding the Brown cluster vocabulary, adding phonological features, and incorporating an improved tokeniser, achieving **93.2%** accuracy—the state of the art for several years. However, CRFs offer no principled mechanism for injecting structural token-type signals beyond manually crafted indicator features.

4.3 Neural Approaches

4.3.1 Bi-LSTM and Bi-LSTM-CRF

The Bi-LSTM-CRF architecture (Lample et al., 2016) reads sequences in both directions and enforces global tag consistency through a CRF decoding layer. Character-level CNNs encode subword information, handling OOV tokens without explicit feature engineering. Plank et al. (2016) demonstrated that multi-task learning with auxiliary objectives improves accuracy on noisy Twitter text.

4.3.2 Standard BERT

BERT (Devlin et al., 2019) introduced deeply bidirectional pre-training via masked language modelling on Wikipedia and BookCorpus. Fine-tuning BERT on tweet POS data produces approximately **94.6%** token-level accuracy on journalistic tweet test sets. However, the model’s training distribution is entirely formal text. Crucially, emojis are mapped to [UNK] by the WordPiece tokeniser, losing all semantic content, and capitalised alert markers like **BREAKING** are treated as ordinary proper nouns with no structural weight.

4.3.3 BERTweet

BERTweet (Nguyen et al., 2020) is a RoBERTa-based model pre-trained on 850 million English tweets. By learning representations from in-domain data, BERTweet captures Twitter vocabulary, emoji semantics, and informal syntax far better than standard BERT, achieving approximately **95.4%** accuracy on journalistic tweet POS benchmarks—the strongest generic baseline available.

However, BERTweet exhibits a fundamental architectural limitation: it treats all tokens as *semantically flat text*. There is no structural distinction between a hashtagged proper noun and a regular one, between an emoji serving as an alert marker and one serving as an emotional reaction, or between **BREAKING** and any other capitalised word. The model must infer all such distinctions purely from contextual co-occurrence statistics—a task for which a model pre-trained on general tweets is poorly calibrated for the breaking-news register.

4.3.4 The Research Gap: Missing Structural Metadata Injection

Surveying the literature from rule-based systems through to BERTweet, a consistent limitation emerges: **no existing approach natively injects structural token-type metadata directly into the transformer’s hidden state during the forward pass**. Prior systems either engineer token-type features as inputs to a CRF or ignore them entirely. The transformer architecture already uses additive positional embeddings—demonstrating a clear precedent for injecting metadata through addition. Projecting token-type embeddings into the model’s 768-dimensional hidden space and adding them directly to token representations is mathematically aligned with this design. This insight has not been exploited in the existing tweet POS tagging literature, constituting the central research gap this work addresses.

4.3.5 Few-Shot Approaches with Large Language Models

Very large generative models (GPT-3, GPT-4, LLaMA) can perform POS tagging via in-context prompting without task-specific fine-tuning. Preliminary studies show competitive zero-shot accuracy, though fine-tuned domain-adapted models remain stronger on standard benchmarks. For the journalistic domain, models with explicit structural awareness represent the more principled choice.

5 Evaluation Methodology

5.1 Token-Level Accuracy

The dominant metric is token-level accuracy: the proportion of tokens assigned the correct tag. For journalistic tweets, accuracy on news-specific token classes (hashtags, alert markers, emoji) is of particular interest and should be reported separately alongside overall accuracy.

5.2 Out-of-Vocabulary Analysis

Given the high OOV rate in tweet data, it is standard to report accuracy separately for in-vocabulary (IV) and out-of-vocabulary (OOV) tokens. Systems relying on word-level features tend to show a large IV/OOV gap, while character-level neural models and subword tokenisers reduce this. For the journalistic domain, compound hashtags are a persistent source of OOV tokens.

5.3 Cross-Domain and Temporal Evaluation

Fromreide et al. (2014) note that models performing well on one Twitter subcorpus can degrade significantly on another. Cross-domain evaluation is essential for assessing real-world robustness.

5.4 Statistical Significance Testing

Comparisons between systems should be accompanied by significance testing. The McNemar test is commonly used for paired comparisons. Unfortunately, many published papers report raw accuracy without significance tests, making it difficult to determine whether small performance differences are meaningful.

6 Comparative Analysis of Systems

Table 2 summarises the reported accuracy of representative systems on journalistic tweet test sets, contextualising the literature and locating the structural metadata gap.

Three trends are clearly visible. First, domain-specific features provide substantial gains over general-purpose taggers even within the CRF paradigm. Second, the transition to neural models brought incremental improvements through automatic feature learning. Third, BERT-family models produce the largest single gains—but even BERTweet, the strongest generic model, leaves a structurally motivated gap. The consistent pattern across all reviewed systems is that none inject token-type structural metadata into the transformer’s hidden state.

Table 2: Comparative accuracy on journalistic tweet test sets. All figures are approximate token-level accuracy percentages.

System	Year	Key Limitation for News Tweets	Acc. (%)
Stanford Tagger (off-shelf)	2003	No Twitter features; OOV collapse	~83.0
Gimpel et al.	2011	Hand-crafted features; no emoji handling	89.4
Owoputi et al.	2013	Cluster-based; no structural injection	93.2
Plank et al.	2016	No news-domain metadata	93.6
Standard BERT	2019	Emoji as [UNK]; no domain adapt	~94.6
BERTweet (fine-tuned)	2020	Flat token treatment; no structure	~95.4
<i>News-Aware (Proposed)</i>	—	<i>Gap addressed by design</i>	>95.8

7 Open Challenges and Future Directions

7.1 Structural Metadata Integration

As identified in Section 4.3.4, no existing tweet POS tagger natively incorporates structural token-type metadata into the transformer’s representational pathway. The design space includes choices of token-type taxonomy (how many categories, how they are defined), projection strategy (addition vs. concatenation), and normalisation scheme. The key insight from the literature on positional embeddings is that *addition* is architecturally aligned with how transformers already integrate metadata, while concatenation introduces dimensionality mismatch and acts as noise against the pre-trained weights.

7.2 Emoji as Syntactic Actors

The NLP community has largely treated emoji as sentiment or topic signals rather than syntactic entities. In the journalistic domain, emoji frequently occupy definite syntactic positions: an alert emoji functions as an interjection; a downward-arrow emoji functions as a verb-equivalent predicate. Semantic translation—mapping emoji to conceptual text before tokenisation—is a promising preprocessing strategy, but its interaction with downstream model behaviour remains an open research question.

7.3 Handling Temporal Vocabulary Drift

Twitter vocabulary evolves rapidly. Neologisms, trending hashtags, and new slang emerge and recede on timescales of days or weeks. Continuous learning approaches—incrementally updating model parameters on new unlabelled data—are promising but require careful

management of catastrophic forgetting.

7.4 Robustness to Demographic Variation

Blodgett et al. (2017) showed that NLP systems exhibit lower accuracy on tweets written in AAVE. Building taggers that perform equitably across demographic groups requires both diverse annotated data and evaluation protocols that measure demographic disparities explicitly.

7.5 Efficient Neural Architectures

State-of-the-art Transformer-based taggers are computationally expensive. Deploying them in real-time breaking-news monitoring pipelines demands efficient inference. Knowledge distillation, quantisation, and lightweight structural embedding modules added on top of a distilled backbone may offer a favourable accuracy-efficiency trade-off.

8 Conclusion

This survey has reviewed the landscape of part-of-speech tagging for Twitter and social media text, with a focused lens on the journalistic breaking-news register. Progress has been substantial: accuracy on standard English benchmarks has risen from around 83% for off-the-shelf taggers to over 95% for fine-tuned domain-adapted Transformers such as BERTweet.

Yet the survey reveals a consistent architectural gap: no existing system natively injects structural token-type metadata into the transformer’s hidden state. This gap is particularly consequential for journalistic tweets, where hashtags, alert markers, and emoji carry disproportionate syntactic and semantic weight. The transformer architecture’s use of additive positional embeddings suggests a natural mechanism for addressing this gap—projecting token-type embeddings into the model’s hidden dimension and adding them to token representations before classification.

The present survey provides the theoretical and empirical foundation for a news-aware POS tagger that exploits this mechanism, situating the proposed architectural innovation within the existing literature and demonstrating that the research gap it addresses is both real and unresolved.

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